

Genes, Chromosomes and DNA

The diagram illustrates the hierarchy of genetic material. At the top left is a green cell with a black nucleus. A dashed line connects the nucleus to a box labeled 'The human cell nucleus contains ____ pairs of chromosomes.' To the right is a purple chromosome with a centromere and sister chromatids. A dashed line connects the chromosome to a box labeled 'Chromosomes are always in _____. There is one chromosome from each _____.' Below the chromosome is a box labeled 'Chromosomes carry _____. Different genes _____ the development of different types of _____, e.g. hair and eye colour.' At the bottom is a DNA double helix with a label 'Base pair' pointing to one of the rungs. A dashed line connects the DNA to a box labeled 'The DNA is _____ up to form the _____ of the _____.' Another dashed line connects the DNA to a box labeled 'A _____ is a short length of the chromosome...'. Below this is a box labeled '...which is quite a long length of _____.' At the bottom right is a box labeled 'There can be _____ of the same gene, which give different versions of a _____, like blue or brown eyes. The different versions of the same gene are called _____ instead of genes.'

Most cells in your body have a nucleus. The nucleus contains your _____ material in the form of _____.

The human cell nucleus contains ____ pairs of chromosomes.

Chromosomes are always in _____. There is one chromosome from each _____.

Chromosomes carry _____. Different genes _____ the development of different types of _____, e.g. hair and eye colour.

The DNA is _____ up to form the _____ of the _____.

A _____ is a short length of the chromosome...

...which is quite a long length of _____.

There can be _____ of the same gene, which give different versions of a _____, like blue or brown eyes. The different versions of the same gene are called _____ instead of genes.

Use the words below to fill in the gaps and boxes above:

DNA molecule, chromosome, parents, arms, versions, gene, alleles, different, characteristic, gene, DNA, coiled, chromosome, genes, control, characteristics, 23, genetic, chromosomes, pairs.